

AIoT Homework (Reinforcement Learning Phase)

Due: May 27, 2026

Following the question given in previous exam, you are asked to implement the MDP according to your answer in the exam) and implement the solution using the Q-Learning algorithm.

❖ Scenario:

Consider the following Scenario: AI-Powered Smart Greenhouse Management System

You are tasked with designing an intelligent agent for a Smart Greenhouse that grows high-value medicinal plants. The greenhouse is equipped with various IoT sensors (measuring soil moisture, ambient temperature, and sunlight) and actuators (a water pump, a heating system, and motorized shade curtains).

The plants are sensitive: if the soil gets too dry, they wilt (reducing yield); if the temperature fluctuates too much, they become diseased. However, the greenhouse must also operate efficiently, water is scarce, and electricity for heating is expensive during peak hours.

Question: Designing the MDP Framework

Drawing on the mathematical formulation of an MDP, $MDP(S, A, T, R, \gamma, H)$, provide a detailed design for this system. Your response must include the following:

- ❖ State Space (S): Define at least three parameters that constitute the environment's state. Specify whether these are discrete or continuous and explain why.
- ❖ Action Space (A): Define the set of possible actions the agent can take through its actuators. Are these actions dependent on the current state?
- ❖ Reward Function (R): Design a scalar reward function that aligns with the agent's objective of "goal-directed learning". How will you balance the reward for plant health against the "negative rewards" (costs) of water and electricity usage?
- ❖ Transition Function (T): *Describe the "Transition model" for this scenario.* For example, if the agent turns on the heater, what is the probability that the state will transition to a specific higher temperature range in the next time step?
- ❖ RL Objective and Solution:
 - State the mathematical objective of the agent in terms of maximizing rewards over time.
 - Propose which RL algorithm (e.g., Value Iteration, Policy Iteration, or Q-Learning) would be most suitable to find the Optimal Policy (π^*) if the environment is highly stochastic (unpredictable weather) and justify your choice.

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❖ Core State Variables

Parameter	Type (suggested)	Lower Bound	Upper Bound	Rationale
Soil Moisture (%)	Continuous	0%	100%	Physical limits of soil saturation
Temperature (°C)	Continuous	5°C	45°C	Below 5°C plants die; above 45°C extreme stress
Sunlight Intensity (W/m ²)	Continuous	0	1200	Typical solar radiation range
Time of Day (hour)	Discrete	0	23	Captures diurnal patterns
Soil Moisture Trend	Continuous	-10%/step	+10%/step	Captures drying/watering dynamics
Temperature Trend	Continuous	-5°C/step	+5°C/step	Captures heating/cooling inertia

- Note: Minimum requirement (as per given exam question): you can use [Soil Moisture \(%\)](#) , [Temperature \(°C\)](#) , and [Sunlight Intensity \(W/m²\)](#) , but adding trends improves realism.
- Note: you should upload your homework source code, and the readme file (text file) by due date.
- Each student should present his/her demo in the class in Wednesday May 27, 2026.

❖ Suggested Discretization (for Q-learning implementation)

Variable	Bin
Soil Moisture	{Dry (0–30), Optimal (30–70), Wet (70–100)}
Temperature	{Low (5–18), Optimal (18–28), High (28–45)}
Sunlight	{Low (0–300), Medium (300–700), High (700–1200)}

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❖ **Action Space**

Action	Description
a_0	Do nothing
a_1	Turn ON water pump
a_2	Turn OFF water pump
a_3	Turn ON heater
a_4	Turn OFF heater
a_5	Open shade curtains
a_6	Close shade curtains

❖ **Reward Function (R)**

You are expected to define the reward function according to your exam's answer.

❖ **Define Optimal Ranges**

Parameter	Optimal Range
Soil Moisture	40–70%
Temperature	20–28°C
Sunlight	300–800 W/m ²